Dataset Details & Documentation: Ames Housing Dataset & Synthetic Augmentation

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Domain: Real Estate & PropTech Valuation Intelligence

Capstone: Prodapt AI/ML Engineering Capstone

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1. Dataset Overview

- **Dataset Name**: Ames Housing Dataset with Gaussian Copula Synthetic Augmentation
- **Primary Source**: OpenML Data Repository (Dataset ID: 43926)
- **Primary Link**: [OpenML Ames Housing Dataset](https://www.openml.org/search?type=data&sort;=runs&id;=43926)
- **Original Author**: Dean De Cock (Truman State University, 2011)
- **Total Record Count**: **10,000 records**
- **Real Historical Transactions**: **2,930 records** (IDs: `PROP-0001` through `PROP-2930`)
- **Synthetic Augmented Transactions**: **7,070 records** (IDs: `SYN-00001` through `SYN-07070`)
- **Total Neighborhoods**: **28 distinct Ames neighborhoods**
- **Construction Year Range**: **1872 to 2010**
- **Standardized Clean File**: `data/sample_data/ames_housing_cleaned.csv`
- **Dense Embedding Matrix**: `data/sample_data/property_embeddings.npy` (10,000 x 384 matrix)

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2. Dataset Purpose & Role in System

In this valuation intelligence system, the dataset serves as the foundational data source for:

1. **Semantic & Hybrid Property Retrieval**: All 10,000 properties are serialized into natural language documents for dense vector indexing and TF-IDF keyword searching.
2. **Comparable-Property Ground Truth**: Only genuine historical sales (`is_synthetic == False`) are retrieved as comparable properties for client reports, ensuring grounded market evidence.
3. **Supervised ML Price Modeling**: Training a `GradientBoostingRegressor` on the augmented training set (n = 9,414), validated against a held-out test set of real transactions (n = 586) achieving R^2 = 0.9198.
4. **Physical & Market Risk Scoring**: Computing neighborhood price volatility, age depreciation, and layout liquidity benchmarks.
5. **Continuous Calibration Baseline**: Providing historical price distribution anchors for the appraiser feedback loop.

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3. The 15 Standardized Key Fields

The system extracts, validates, and models the following 15 key features across all real and synthetic records:

Key Field	Data Type	Physical Description	Sample Range / Values
SalePrice	Numeric (USD)	Historical property transaction closing price (Target variable).	\$12,789 – \$755,000 (Median: \$160,000)
OverallQual	Integer (1–10)	Rates the overall material quality and architectural finishes.	1 (Very Poor) to 10 (Very Excellent)
OverallCond	Integer (1–10)	Rates the current physical maintenance and condition.	1 (Very Poor) to 10 (Very Excellent)

Key Field	Data Type	Physical Description	Sample Range / Values
GrLivArea	Numeric (sq ft)	Above-grade (ground level) finished living area.	334 – 5,642 sq ft
TotalBsmtSF	Numeric (sq ft)	Total square footage of basement area.	0 – 6,110 sq ft
GarageCars	Integer	Size of garage in terms of car capacity.	0 – 5 cars
GarageArea	Numeric (sq ft)	Size of garage in square feet.	0 – 1,488 sq ft
YearBuilt	Integer (Year)	Original construction year.	1872 – 2010
YearRemodAdd	Integer (Year)	Remodel date (same as construction year if no remodel).	1950 – 2010
FullBath	Integer	Full bathrooms above grade.	0 – 4 bathrooms
BedroomAbvGr	Integer	Number of bedrooms above basement level.	0 – 8 bedrooms
TotRmsAbvGrd	Integer	Total rooms above ground (excluding bathrooms).	2 – 15 rooms
Neighborhood	Categorical	Location / subdivision within Ames (28 distinct neighborhoods).	North Ames, College Creek, Stone Brook, Somerset, Edwards, etc.
LotArea	Numeric (sq ft)	Lot size in square feet.	1,300 – 215,245 sq ft
GarageYrBlt	Integer (Year)	Year garage was constructed (imputed from YearBuilt if null).	1895 – 2010

4. Synthetic Data Generation Methodology

To enrich training diversity while preserving statistical validity, 7,070 synthetic records were generated using a neighborhood-conditioned Gaussian Copula framework (`generate_synthetic_data.py`):

1. **Neighborhood Conditioning**: Separate Gaussian Copula models were fitted for each of the 28 neighborhoods, conditioning distributions on local price levels and architectural styles.
2. **Empirical Marginals**: Continuous features (square footage, lot size, prices) were mapped to empirical cumulative distribution functions (ECDF) to preserve skewness and multi-modality.
3. **Novelty Distance Filter**: Synthetic candidates were evaluated against their nearest real neighbor using Euclidean distance across standardized features. Candidates falling closer than the 25th percentile of real nearest-neighbor gaps were rejected to prevent memorization and duplicate points.
4. **Physical Integrity Constraints**: Synthetic records enforce physical relationships:

• `YearRemodAdd >= YearBuilt`

• `TotalBsmtSF >= 0` and `GarageArea >= GarageCars * 150`

• Standard 1–10 integer constraints on `OverallQual` and `OverallCond`

5. **Metadata Tagging**: All synthetic records are flagged with `is_synthetic = True` and assigned IDs starting with `SYN-`.

5. Honest Train / Test Split & Comparable Protection

To uphold rigorous scientific standards and avoid synthetic data leakage:

1. **Real-Only Test Evaluation**:

• The test set (n = 586) consists strictly of a 20% random sample (`random_state=42`) of the **2,930 real historical records**.

• **Synthetic records never appear in the test set**.

2. Training Set Composition:

• The training set contains 2,344 real records (80% of real) plus all 7,070 synthetic records, totaling **9,414 training rows**.

3. Comparable Sales Isolation:

• In all comparable property search workflows, synthetic records are filtered out by default (`include_synthetic = False`). Real estate clients and appraisers are presented only with genuine historical sales transactions.

Model Evaluation on Real Test Set (n = 586):

• **Trained Model (Real + Synthetic)**: R^2 = 0.9198, RMSE = \25,361.84$, MAE = \16,161.76$, MAPE = 8.87\%

• **Baseline Model (Real Only)**: R^2 = 0.9031, RMSE = \27,870.39$, MAE = \15,889.09$, MAPE = 8.46\%

Synthetic augmentation provided a **+0.0167 gain in R^2** and reduced RMSE by **\$2,508.55** on real unseen test transactions.

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6. Derived Features & Searchable Representation

During data loading and preprocessing, the pipeline creates:

• **PropertyAge**: `2026 - YearBuilt`

• **EffectiveAge**: `2026 - YearRemodAdd`

• **PricePerSqFt**: `SalePrice / GrLivArea`

• **searchable_text**: Textual document summarizing location, layout, quality, and historical sales metrics (1 row = 1 document, no text chunking required).